

water pollution

what is water pollution?

Water pollution is any change in water quality that can harm living organisms or make the water unfit for human uses such as irrigation and recreation. It usually involves contamination by one or more chemicals or excessive heat (thermal pollution).

water pollution sources

- Point sources
- Non-point sources

water pollution sources

Point sources (single source):

- **Point sources** discharge pollutants into bodies of surface water at specific locations through drain pipes, ditches, or sewer lines.
- Point sources are located at specific places, they are fairly easy to identify, monitor, and regulate.



Examples include factories, sewage treatment plants (which remove some, but not all, pollutants), underground mines, and oil tankers.

Figure 20-3 This point-source water pollution flows uncontrolled into a stream near Gargas, France.

water pollution sources

Nonpoint sources:

- **Nonpoint sources** are broad and diffuse areas, rather than points, from which pollutants enter bodies of surface water or air.
- We have made little progress in controlling water pollution from nonpoint sources because of the difficulty and expense of identifying and controlling discharges from so many diffuse sources.



Examples include runoff of chemicals and sediments from cropland, livestock feedlots, logged forests, urban streets, parking lots, lawns, and golf courses.

Water pollution causes

- Agricultural activities
- Industrial facilities
- Mining
- Plastics

water pollution causes

- *Agricultural activities* are by far the leading cause of water pollution. Sediment eroded from agricultural lands is the most common pollutant.
- Other major agricultural pollutants are
fertilizers and pesticides
bacteria from livestock
food-processing wastes
excess salts from soils of irrigated cropland.

water pollution causes

- Industrial facilities, which emit a variety of harmful inorganic and organic chemicals, are a second major source of water pollution.

Coal industry contribution:

A growing problem of great concern is that of coal ash, an indestructible waste created by the burning of coal in power plants. Some power utilities store the ash on land in slurry ponds that can leak and pollute nearby bodies of water. Other utilities dump the toxic ash and associated wastewater into nearby lakes and rivers.

Oil industry contribution:

In addition, some oil and natural gas drilling operations inject large volumes of water at high pressure into shale rock deposits to release trapped natural gas or oil. This process can contaminate aquifers that are used as sources of drinking water with various chemicals.

water pollution causes

- Mining

Mining is the third biggest source of water pollution. Surface mining disturbs the land creating major erosion of sediments and runoff of toxic chemicals.

Wastes from existing and abandoned mines can also pollute streams and lakes.



water pollution causes

- Plastics

water pollution is caused by the widespread use of human-made materials such as plastics that make up millions of products, all of which eventually end up in the environment.

The polymers that make up the plastics break down very slowly and, in the process, pollute many waterways and the oceans where they have been improperly discarded



Effects of water pollution

Major Water Pollutants and Their Sources

Type/Effects	Examples	Major Sources
Infectious agents (pathogens) Cause diseases	Bacteria, viruses, protozoa, parasites	Human and animal wastes
Oxygen-demanding wastes deplete dissolved oxygen needed by aquatic species	Biodegradable animal wastes and plant debris	Sewage, animal feedlots, food-processing facilities, paper mills
Plant nutrients Cause excessive growth of algae and other species	Nitrates (NO_3^-) and phosphates (PO_4^{3-})	Sewage, animal wastes, inorganic fertilizers
Organic chemicals Add toxins to aquatic systems	Oil, gasoline, plastics, pesticides, fertilizers, cleaning solvents	Industry, farms, households, mining sites, runoff from streets and parking lots
Inorganic chemicals Add toxins to aquatic systems	Acids, bases, salts, metal compounds	Industry, households, mining sites, runoff from streets and parking lots

Effects of water pollution

Major Water Pollutants and Their Sources

Type/Effects	Examples	Major Sources
Sediments Disrupt photosynthesis, food webs, other processes	Soil, silt	Land erosion from farms and construction and mining sites
Heavy metals Cause cancer, disrupt immune and endocrine systems	Lead, mercury, arsenic	Unlined landfills, household chemicals, mining refuse, industrial discharges
Thermal Make some species vulnerable to disease	Heat	Electric power and industrial plants

Water pollution problems

- Streams and lakes
- Ground water
- Oceans

Water pollution problems in streams and lakes

Flowing rivers and streams can recover rapidly from **moderate levels** of degradable, oxygen-demanding wastes through a combination of dilution and bacterial breakdown of such wastes.

But this process does not work when streams become **overloaded** with such pollutants or when drought, damming, or water diversion reduce their flows.

This process (dilution and bacterial breakdown) does not eliminate slowly degrading or nondegradable wastes.

Water pollution problems in streams and lakes

Lakes and Reservoirs Vulnerable to Water Pollution

- Lakes and reservoirs are generally less effective at diluting pollutants than streams are, for two reasons.
- *First*, lakes and reservoirs often contain stratified layers that undergo little vertical mixing.
- *Second*, they have little or no flow.
- The flushing and changing of water in lakes and large artificial reservoirs can take from 1 to 100 years, compared with several days to several weeks for streams.
- As a result, lakes and reservoirs are more vulnerable than streams are to contamination by runoff or discharge of plant nutrients, oil, pesticides, and non-degradable toxic substances, such as lead and mercury.

Water pollution problems in streams and lakes

Cultural Eutrophication

- Eutrophication is the name given to the natural nutrient enrichment of a shallow lake, estuary, or slow moving stream. It is caused mostly by runoff of plant nutrients such as nitrates and phosphates from surrounding land.
- Near urban or agricultural areas, human activities can greatly accelerate the input of plant nutrients to a lake—a process called **cultural eutrophication**.
- Sources for cultural eutrophication
farms,
feedlots,
chemically fertilized lawns,
mining sites,
sewage treatment plants.

Water pollution problems in streams and lakes

Cultural Eutrophication-effects

Effects fish growth by disturbing photosynthesis

- Nutrient overload produces dense growths of organisms such as algae and cyanobacteria and thick growths of water hyacinth duckweed, and other aquatic plants. These dense colonies of plant life can reduce lake productivity and fish growth by decreasing the input of solar energy needed for photosynthesis by the phytoplankton that support fish populations



Water pollution problems in streams and lakes

Cultural Eutrophication-effects

Depletes oxygen levels in water

- When the algae die, they are decomposed by swelling populations of aerobic bacteria, which deplete dissolved oxygen in the surface layer of water near the shore as well as in the bottom layer of a lake. This can kill fish and other aerobic aquatic animals;



Major Pollution Problems Affecting Groundwater

- According to many scientists, groundwater pollution is a serious threat to human health.

Common **pollutants** that can contaminate ground water are

fertilizers

pesticides

gasoline (or crude oil) spills,

Paint thinners

organic solvents

Major Pollution Problems Affecting Groundwater

- When groundwater becomes contaminated, it **cannot cleanse itself** of degradable wastes as quickly as flowing surface water can.

Reasons

- (a) Groundwater flows so slowly—usually less than 0.3 meter (1 foot) per day—that contaminants are not diluted and dispersed effectively.
- (b) Groundwater usually has much lower concentrations of dissolved oxygen (which helps decompose many contaminants) and smaller populations of decomposing bacteria.
- (c) The usually cold temperatures of groundwater also slow down chemical reactions that decompose wastes.

Major Pollution Problems Affecting Groundwater

- Ground pollution is a serious threat

Nitrate and nitrite ions contamination

- In agricultural areas where nitrates in fertilizer can leach into groundwater. Nitrite ions (NO_2^-) in the stomach, colon, and bladder can convert some of the nitrate ions in drinking water to organic compounds that have been shown in tests to cause cancer in more than 40 animal species.
- The conversion of nitrates in tap water to nitrites in infants under 6 months old can cause a potentially fatal condition known as "blue baby syndrome," in which blood lacks the ability to carry sufficient oxygen to body cells.

Major Pollution Problems Affecting Groundwater

- Ground pollution is a serious threat

Arsenic contamination

- According to a 2007 study by the WHO, more than 140 million people in 70 countries are drinking water with arsenic concentrations of 5-100 times the accepted safe level of 10 parts per billion (ppb).
- The WHO estimates that long-term exposure to nondegradable arsenic in drinking water is likely to cause hundreds of thousands of premature deaths from cancer of the skin, bladder, and lung.

Major Pollution Problems Affecting Oceans

Ocean Pollution Is a Growing and Poorly Understood Problem

Coastal ocean water pollution: (viruses and bacteria growth)

- The UN Environment Programme (UNEP) report says that 80-90% of the municipal sewage from most coastal areas of less-developed countries, and in some such areas of more-developed countries, is dumped into oceans without treatment.
- Recent studies of some U.S. coastal waters have found vast colonies of viruses thriving in raw sewage and in effluents from sewage treatment plants (which do not remove viruses) and leaking septic tanks.

According to one study, one-fourth of the people using coastal beaches in the United States develop ear infections, sore throats, eye irritations, respiratory disease, or gastrointestinal disease from swimming in seawater containing infectious viruses and bacteria.

Major Pollution Problems Affecting Oceans

Ocean Pollution Is a Growing and Poorly Understood Problem

Coastal ocean water pollution: (harmful algae growth)

- Runoffs of sewage and agricultural wastes into coastal waters introduce large quantities of nitrate(NO_3^-) and phosphate (PO_4^{3-}) plant nutrients, which can cause explosive growths of **harmful algae**.
- **Contamination of seafood:** Harmful algae can release waterborne and airborne toxins that poison seafood, damage fisheries, kill some fish-eating birds.
- **oxygen depleted zones:** When the algae die, they are decomposed by colonies of oxygen-consuming bacteria. Evidence indicates that oxygen-depleted zones are caused mostly by excessive inputs of nitrates and phosphates from runoff of fertilizers and animal wastes and by deposition of nitrogen compounds from the atmosphere.

Major Pollution Problems Affecting Oceans

Ocean Pollution Is a Growing and Poorly Understood Problem

Ocean Pollution from Oil

Crude petroleum and refined petroleum reach the ocean from a number of sources and become highly disruptive pollutants

- Tanker accidents and blowouts at offshore drilling rigs
- The largest source of ocean pollution from oil is urban and industrial runoff from land

At least 37% of the oil reaching the oceans is waste oil, dumped, spilled, or leaked onto the land or into sewers by cities and industries,

Dealing with water pollution

Reducing Surface Water Pollution from Nonpoint Sources

- Farmers can reduce soil erosion by keeping cropland covered with vegetation and using other **soil conservation** methods.
- They can also **reduce the amount of fertilizer** that runs off into surface waters by using slow-release fertilizer, using no fertilizer on steeply sloped land, and planting buffer zones of vegetation between cultivated fields and nearby surface waters.
- **Organic farming** can also help to prevent water pollution caused by nutrient overload because it does not use commercial inorganic fertilizers and pesticides.
- By applying pesticides only when needed and relying more on **integrated pest management** farmers can also reduce pesticide runoff.
- control runoff and infiltration of manure from animal feedlots by planting buffer zones and by locating feedlots, pastures, and animal waste storage sites away from steeply sloped land, surface water, and flood zones

Dealing with water pollution

Sewage Treatment

In developed countries, raw sewage reaching a treatment plant typically undergoes one or two levels of wastewater treatment.

The first is **primary sewage treatment**—a *physical* process that uses screens and a grit tank to remove large floating objects and to allow solids such as sand and rock to settle out. Then the waste stream flows into a primary settling tank where suspended solids settle out as sludge.

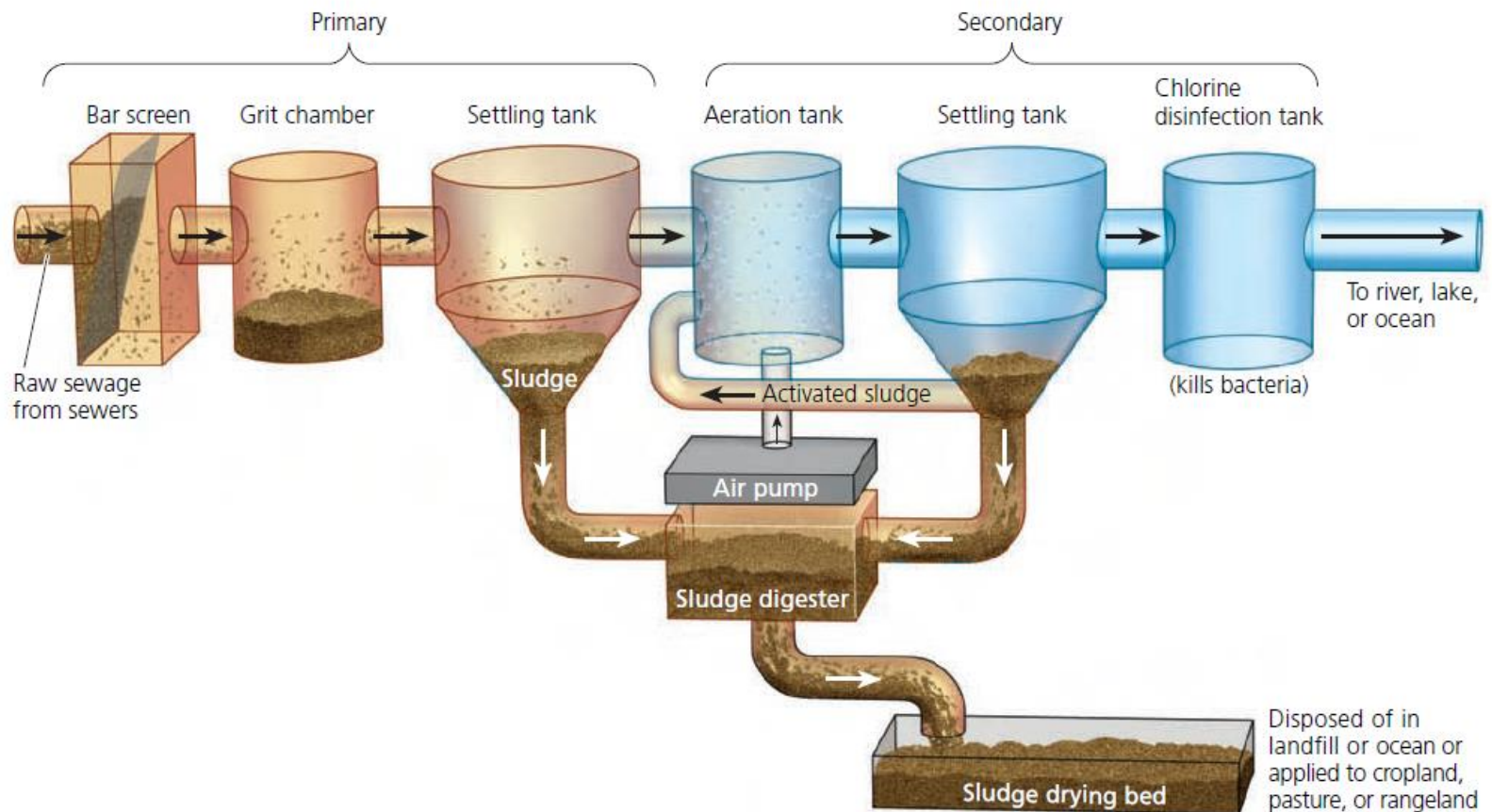
The second level is **secondary sewage treatment**— a biological process in which aerobic bacteria remove as much as 90% of dissolved and biodegradable, oxygen-demanding organic wastes

A third level of clean-up, **advanced or tertiary sewage treatment**, uses a series of specialized chemical and physical processes to remove specific pollutants. In its most common form, advanced sewage treatment uses special filters to remove phosphates and nitrates from wastewater before it is discharged into surface waters

Dealing with water pollution

Sewage Treatment

water from sewage treatment plants usually undergoes **bleaching**, to remove water coloration, and disinfection to kill disease-carrying bacteria and some (but not all) viruses. The usual method for accomplishing this is **chlorination**.



Dealing with water pollution

Ways to Reduce and Prevent Water Pollution

- Prevent groundwater contamination
- Reduce nonpoint runoff
- Reuse treated wastewater for drinking and irrigation
- Find substitutes for toxic pollutants
- Reduce air pollution
- Reduce poverty
- Slow population growth
- Practice the three R's of resource use (reduce, reuse, recycle)

Dealing with water pollution

Individual role in prevention of water pollution

- Fertilize garden and yard plants with manure or compost instead of commercial inorganic fertilizer
- Minimize your use of pesticides, especially near bodies of water
- Prevent yard wastes from entering storm drains
- Do not flush unwanted medicines down the toilet
- Do not pour pesticides, paints, solvents, oil, antifreeze, or other products containing harmful chemicals down the drain or onto the ground

Hazards
due to air and water pollution

Environmental hazards

- **A hazard** is any agent that can cause harm or damage to humans, property, or the environment.
- **A risk** is the probability of suffering harm from a hazard that can cause injury, disease, death, economic loss, or damage
- **Risk assessment** is the process of using statistical methods to estimate how much harm a particular hazard can cause to human health or to the environment.
- **Risk management** involves deciding whether or how to reduce a particular risk to a certain level and at what cost.

Environmental hazards

We face different types of environmental hazards

- biological hazards
 - chemical hazards
 - nuclear hazards
 - natural hazards
 - cultural hazards
 - life style hazards
-
- **Physical hazards** such as fire, earthquakes, volcanic eruptions, floods, and storms.
 - **Cultural hazards** such as unsafe working conditions, criminal assault, poverty, etc.
 - **Lifestyle hazards** such as smoking, making poor food choices, drinking too much alcohol, etc.

Environmental hazards

Biological hazards

- infectious disease
- transmissible disease
- non-transmissible disease

Environmental hazards

Biological hazards

An infectious disease is caused when a pathogen such as a bacterium, virus, or parasite invades the body and multiplies in its cells and tissues.

examples: flu, malaria, tuberculosis, and measles.

A *bacterial* disease such as tuberculosis spreads as the bacteria multiply.

A *viral* disease such as flu or HIV spreads as viruses take over a cell's genetic mechanisms to copy themselves

Environmental hazards

Biological hazards

- **transmissible disease** (also called a contagious or communicable disease) is an infectious disease that can be transmitted from one person to another.

Examples are flu, tuberculosis, and measles.

- **non-transmissible disease** is caused by something other than a living organism and does not spread from one person to another.

Examples: cardiovascular (heart and blood vessel) diseases, most cancers, asthma, diabetes, and malnutrition.

Environmental hazards

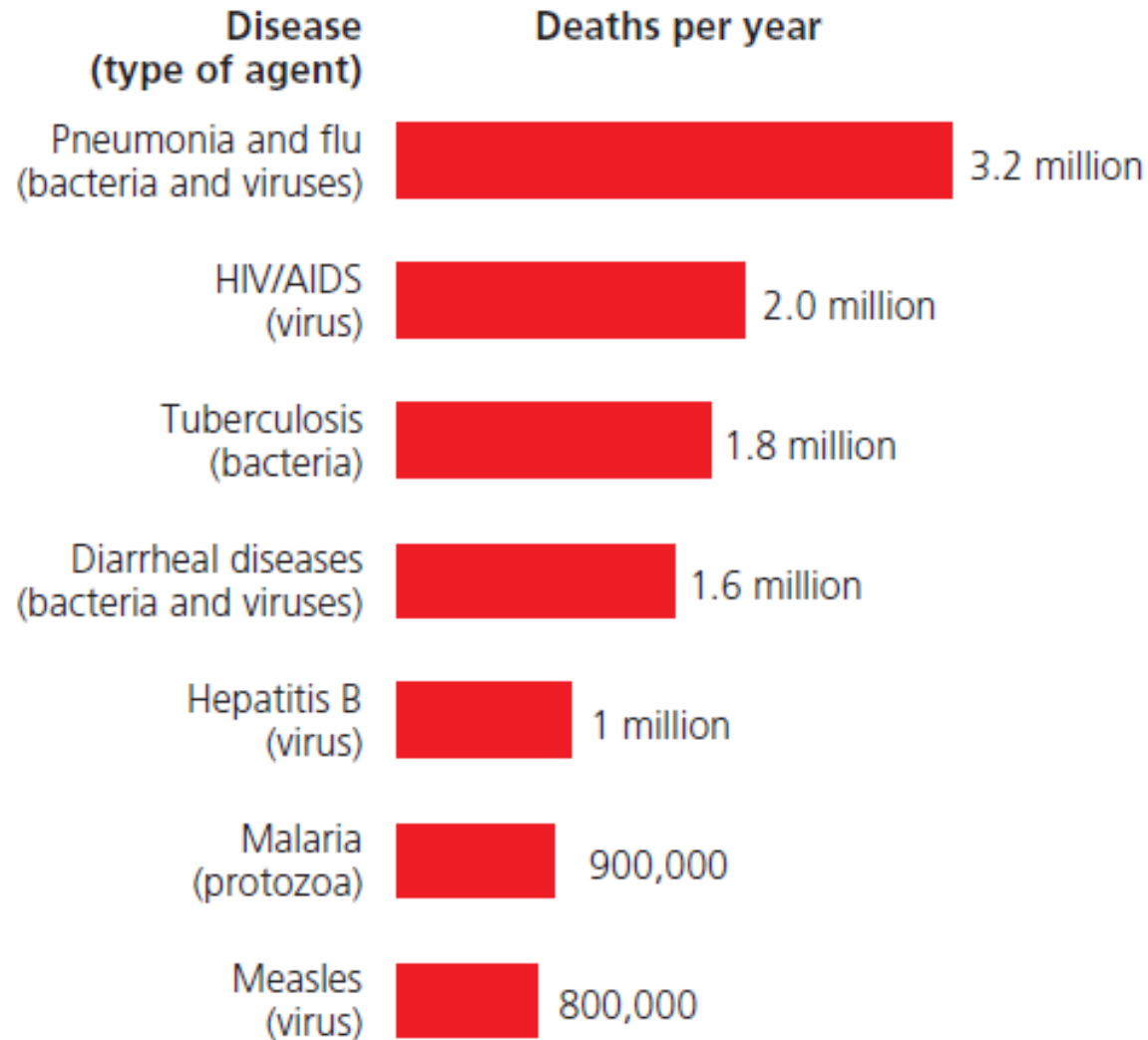
Biological hazards- Infectious Diseases Are Still Major Health Threats

- Infectious diseases remain as serious health threats, especially in less-developed countries.
- A large-scale outbreak of an infectious disease in an area or a country is called an *epidemic*.
- A global epidemic such as tuberculosis or AIDS is called a *pandemic*.
- One reason why infectious disease is still a serious threat is that many disease-carrying bacteria have developed genetic resistance to widely used antibiotics

Example: mosquitoes have become immune to widely used pesticides such as DDT

Environmental hazards

Biological hazards- Infectious Diseases Are Still Major Health Threats



Chemical Hazards

Chemical Hazards

A **toxic chemical** is one that can cause temporary or permanent harm or death to humans and animals.

In 2004, the EPA listed

- arsenic
- lead
- mercury
- vinyl chloride (used to make PVC plastics),
- polychlorinated biphenyls (PCBs)

as the top **five toxic substances** in terms of human and environmental health.

Chemical Hazards

A **toxic chemical** is one that can cause temporary or permanent harm or death to humans and animals.

There are three major types of potentially toxic agents.

- Carcinogens
- mutagens
- teratogens

Chemical Hazards

toxic chemical-carcinogens

Carcinogens are chemicals, types of radiation, or certain viruses that can cause or promote cancer.

Cancer is disease in which malignant cells multiply uncontrollably and create tumors that can damage the body and often lead to premature death.

Examples of carcinogens are arsenic, benzene, formaldehyde, gamma radiation, PCBs, radon, certain chemicals in tobacco smoke, ultraviolet(UV) radiation, vinyl chloride.

Typically, 10-40 years may elapse between the initial exposure to a carcinogen and the appearance of detectable cancer symptoms.

Partly because of this time lag, many healthy teenagers and young adults have trouble believing that their smoking, drinking, eating, and other habits today could lead to some form of cancer before they reach age 50.

Chemical Hazards

toxic chemical-mutagens

Mutagens are chemicals or forms of radiation that cause or increase the frequency of mutations, or changes, in the DNA molecules found in cells.

Most mutations cause no harm but some can lead to cancers and other disorders.

For example,

- nitrous acid (HNO_2), formed by the digestion of nitrite (NO_2^-) preservatives in foods, can cause mutations linked to increases in stomach cancer in people who consume large amounts of processed foods and wine containing such preservatives.

3. Teratogens are chemicals that cause **harm or birth defects** to a foetus or embryo.

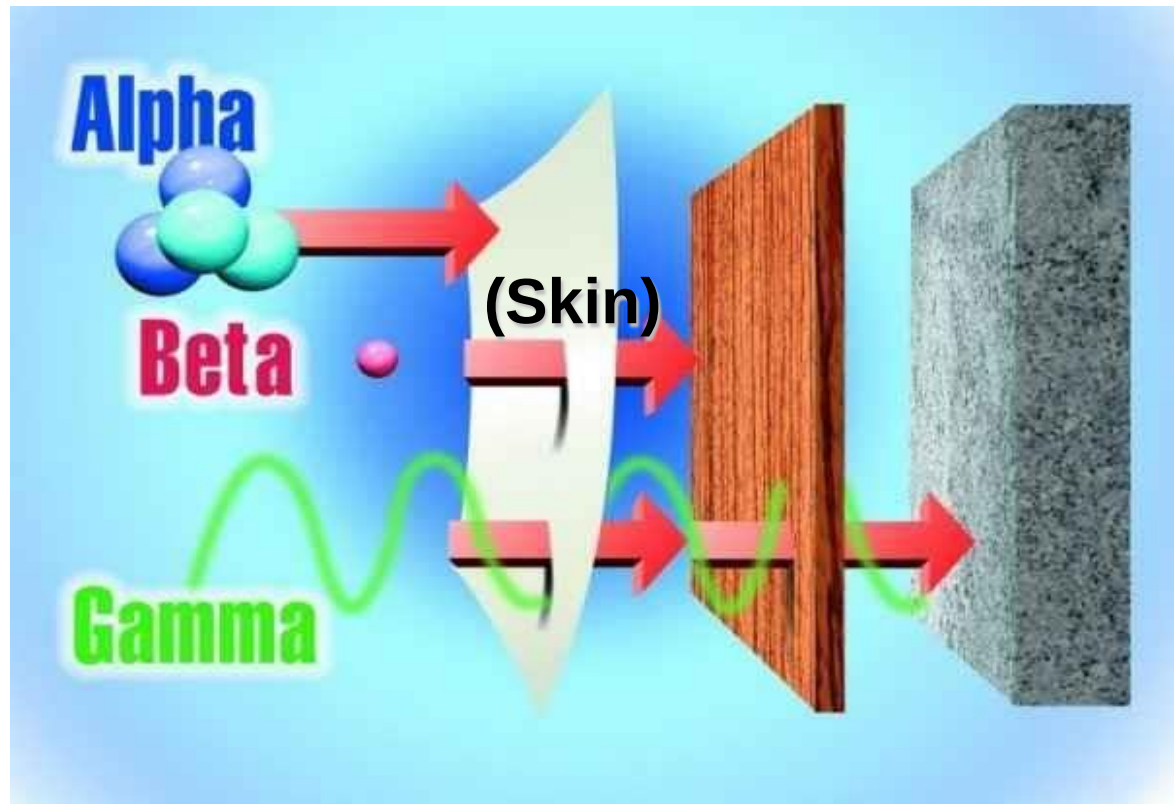
Ethyl alcohol is a teratogen. Drinking during pregnancy can lead to offspring with low birth weight and a number of physical, developmental, behavioural, and mental problems.

Other teratogens are angel dust, benzene, cadmium, formaldehyde, lead, mercury, mescaline, PCBs, phthalates, thalidomide, and vinyl chloride.

Nuclear Hazards: sources of radiation

- Radioactive substances undergo radioactive decay - unstable atoms to stable atoms.
- Radioactive decay release **energy** along with **Gamma rays** (high energy electromagnetic radiation), **Beta particles** (fast moving, -vely charged), and **Alpha particles** (fast moving, +vely charged).

Penetration
power of ionizing
radiations



Sources of Nuclear Radiation

- Nuclear radiation is the major cause of nuclear hazards, it can be of two types:

(1) Natural sources:

- > Cosmic rays from outer space , radioactive radon-222, soil, rocks, air, water and food which contain one or more radioactive substances .

(2) Anthropogenic sources

- > These sources are nuclear power plants , nuclear accidents , X-rays , diagnostic kits , test laboratories etc., where radioactive substances are used.

Effects of Nuclear Hazards to the environment

> In general, the ill effects produced by nuclear weapons are of two types:

- (i) Somatic
- (ii) Genetic

(1) Somatic effects:

The general body cells of the person exposed to nuclear radiation are affected. This may lead to deadly diseases like cancer and shorten the lives of the people.

(2) Genetic effects:

The genes of the person exposed to nuclear radiation get damaged. The genetic disorders are passed onto the next generations. These effects are more dangerous than the somatic effects.